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STABLECOIN OPERATIONAL RISK · SCENARIO MODELING · TOKEN-ECONOMIC MODELING

74-vector risk taxonomy · 64 testable controls · 624-scenario simulation battery · Thorchain $200M insolvency modeled 13 months ahead · 8 SSRN papers

SUMMARY

Operational risk and scenario-modeling strategist. In Dec 2023 authored a 10-step scenario-modeling memo predicting Thorchain's eventual $200M insolvency in a protocol whose token had peaked at $3.3B market cap; predicted failure mode materialized in Jan 2025 (13 months later); the firm held no exposure. Founding research hire at Borderless Capital (RIA, Dec 2019 to Feb 2026); built quantitative token-economic models forecasting emissions, fee capture, subsidy-to-revenue trajectories, and governance concentration under wide scenario ranges. Now founder of Tokenization Systems (Feb 2026 to present); published Operational Risk in Token Economies (74-vector taxonomy mapped to 64 testable controls and 31 program objectives across a 624-scenario simulation battery; Lido V3 staking-vault production case study) and Adaptive Tokenomics (industrial-control-engineering PID controller; 240-scenario Monte Carlo; ~6x worst-case supply-volatility compression under competitor-shock attack). Contributed eight federal regulatory comment letters across stablecoin and banking dockets. Authored the Operational Risk in Token Economies taxonomy: 74 risk vectors mapped to 64 testable controls and 31 program objectives across a 624-scenario simulation battery; Lido V3 staking-vault case study.

EXPERIENCE

Founder · Tokenization Systems *Feb 2026 – Present*

- Authored an eight-paper SSRN-live research program covering the **Control Layer Intensity Index** (CLII) for gateway concentration, the **Minimum Viable Equivalence Pack** (MVEP) for institutional tokenization diligence, programmable monetary-policy design (**Adaptive Tokenomics**), governance concentration, and operational risk in token economies. Three additional papers in development (B5 Bankruptcy-Priority Misaligned Incentives, A7 Control Layer Theory, SVB Causal DiD); methodology paper in preparation.
- Contributed eight comment letters across a federal regulatory program (OCC bank stablecoin engagement RIN 1557-AF41 plus supplemental control-layer extension; Treasury GENIUS Act state similarity; FDIC payment-stablecoin reserve standards RIN 3064-AG19 and stablecoin-subsidiary issuance approvals RIN 3064-AG20; FinCEN AML/CFT FINCEN-2026-0034; FinCEN-OFAC PPSI sanctions FINCEN-2026-0100; joint banking AML/CFT 91 FR 18304); Routing the Dollar prepared for the Federal Reserve Fifth Conference on the Dollar's International Roles (Jun 2026); Governance Concentration in peer review at Frontiers in Blockchain.
- Active dialogues across U.S. national-security research, major securities exchanges, central securities depositories, global payments messaging infrastructure, tier-1 bank tokenization platforms, and SEC-registered transfer agents on stablecoin, deposit-token, tokenized-deposit, tokenized money-market-fund, tokenized-commodity, and tokenized-equity deployment.

Senior Investment Analyst, Digital Assets · Borderless Capital (RIA; first hire) *Dec 2019 – Feb 2026*

- Predicted Thorchain's **$200M insolvency 13 months in advance**; Borderless held no exposure when the predicted failure mode materialized.
- Identified ~15% misreporting of circulating supply in a top-25 Layer 1 protocol by independently modeling its tokenomics (emission schedule, vesting, burn dynamics); the discrepancy had propagated to CoinMarketCap and CoinGecko, and was corrected after partnering with the protocol's foundation engineers.
- Prevented hostile governance capture on a top-25 blockchain via voting-path simulation under coalition-attack scenarios; surfaced the vulnerability and mitigation route before exploitation.
- Built operational-risk taxonomies for tokenized systems: 74 risk vectors mapped to 64 testable controls and 31 program objectives across a 624-scenario simulation battery (published as **Operational Risk in Token Economies**); Lido V3 staking-vault production case study sets operator collateral by risk-score alignment.
- Built quantitative token-economic models forecasting emissions, fee capture, subsidy-to-revenue trajectories, and governance concentration under wide scenario ranges; supply-equilibrium and slashing dynamics supporting Layer 1 mechanism-design advisory.
- Mapped jurisdictional tax / regulatory advantages of DAT structures over direct crypto holding: Japan's Metaplanet DAT cuts crypto capital-gains tax from up to 55% to 20% (equity rate, further reducible via Japan's IRA structure); Hong Kong's Southbound Connect could give mainland Chinese capital regulated access to crypto banned domestically.
- Built apples-to-apples Multiple on NAV (mNAV) methodology for Digital Asset Treasury Stocks (DATs) and shipped a daily mNAV dashboard sourcing SEC filings and press releases; standardized treatment of non-crypto assets in the numerator across issuers (issuer-reported mNAV figures use inconsistent conventions that undermine cross-DAT comparison).

- Advised 100 portfolio companies 1-on-1 on token mechanism design (stablecoin reserves, governance-token allocation and incentive design, yield wrappers, burn/emission dynamics); standards defined in **Introduction to Tokenomics**.
- Translated regulatory change (GENIUS, CLARITY, Singapore MAS, MiCA, state money-transmitter and trust-charter regimes) into product, reserve, routing, and compliance decisions across U.S., Singapore, Cayman / BVI / Bermuda, and EU; coordinated cross-functional partner work spanning Product / Engineering, Risk (operational-risk taxonomy), and Legal.

SELECTED RESEARCH

- **Operational Risk in Token Economies.** Operational-risk taxonomy (74 risk vectors, 64 testable controls, 31 program objectives) mapped to a 624-scenario battery; Lido V3 staking-vault case study.
- **Adaptive Tokenomics.** Industrial control engineering (PID controller) applied to token-supply emission as programmable monetary-policy mechanism; 240-scenario Monte Carlo simulation; PID emission control cuts worst-case node-count variance from CV 0.544 (static emission) to 0.087 (PID), functioning as downside insurance under competitor-shock attack.
- **Routing the Dollar.** Prepared for Federal Reserve / FRBNY Fifth Conference on the Dollar's International Roles (Jun 2026). Gateway concentration; CLII across 19 gateways with Tier 1 a 56% / 44% Coinbase-Circle duopoly; on Ethereum, transfer volume doubled while unique counterparties fell 25% and cross-gateway connections dropped from 5.8% to 2.6%.
- **Minimum Viable Equivalence Pack.** Ten-category institutional diligence standard; finds bank-chartered stablecoin issuers face ~22x the regulatory capital requirements of money-transmitter-chartered issuers at $40B scale; production-validated by Circle's BUIDL March 27, 2025 incident (23 hours of limited availability).
- **Three Tokenization Strategies.** Each of three strategies satisfies different MVEP categories with measurably different strength (Regulated Collateral Circuit, Multi-Issuer Yield Stack, Tokenized Equity Rails); Securitize as cross-strategy concentration risk.
- **Seven Dollars.** Cross-product framework for major dollar-denominated digital products; CLII / MVEP decoupling; stress propagation through dependency chains (SVB, Circle pool, Paxos).
- Dollar v3. Four digital-dollar objects (payment stablecoins, tokenized deposits, deposit tokens, yield wrappers); typology for inheritance of control environments, backstop expectations, and failure modes; load-bearing for institutional scoping of product shape and counterparty topology.
- **Governance Concentration Beyond Token Allocation.** In peer review at Frontiers in Blockchain. 40-protocol cross-section: initial token allocation does not predict governance concentration; protocol sector does (DePIN significantly more concentrated than DeFi; Cohen's d = 0.96).
- **Who Burns the Tokens?** Burn-mechanism design (carrier contracts vs. subscriptions) drives demand concentration: Helium's carrier-contract model concentrates ~90% of token burns in 5 purchasers (HHI 0.22) despite ~1M hotspots; coalition-power analysis extends standard concentration measures for single-address risk.

TECHNICAL

Methods: policy-shock impact estimation (difference-in-differences), Monte Carlo simulation, PID controller design (industrial control engineering), coalition-power analysis, supply equilibrium modeling, build-versus-partner framework design

Data: Dune Analytics, Nansen Pro, Artemis, DefiLlama, CoinGecko, Helius (Solana), Etherscan, Blockscout (EVM), Spacescope, Token Terminal, Tally (DAO governance), rated.network, Beaconcha.in (validator data), FRED (macro); automated SEC filing and press release monitoring

Stack: Python (pandas, statsmodels, scipy, matplotlib), R (fixest, did, synthdid), SQL (DuneSQL, Trino), JavaScript / TypeScript; 14 custom Python MCP servers wrapping the data sources above plus GitHub; Claude.ai + Claude Code + Cursor with 36-skill library and 17-file living-knowledge architecture; n8n, Apify, Git / GitHub

RECOGNITION

- Uniswap Foundation governance committee (2024)
- ETHDenver pitch judge
- **Introduction to Tokenomics** article publicly endorsed by Binance founder CZ on day of publication

FRAMEWORKS AUTHORED

- Authored CLII (A1), MVEP (A2), MGCPT, TCRD; drafted MVEP-AI (eight-skill suite) and CLII-AI.

EDUCATION

High Point University | B.S.B.A., Business Administration (Magna Cum Laude) · Investment Club President · Lingnan University, Hong Kong (exchange) *2018*